

NTU SPACE DATA CENTER

SDTWIN THERMAL

Software Module Test Report

*Parameter Assembly • Surface Environment • Heat Transfer • Electrothermal Coupling •
FE Benchmarks • End-to-End Runs*

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Abstract	This report defines the verification scope, procedures, reference data and acceptance criteria of the SDTwin thermal module according to its design report, and records the execution results of 16 cases: 13 passed and 3 failed.			
Keywords	lumped thermal network; test cases; finite element benchmark; acceptance criteria			
Change Record				
Revision	Change Request	Date	Author	Description
1.0	Initial issue	2026-10-05		Test framework and comparison plan
2.0	Case execution	2026-10-06		All cases executed on the implemented thermal module; actual results, conclusions and evidence recorded

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1 Overview

1.1 Identification

Platform name: SDTwin.

Module under test: Thermal Module.

Document title: Test Report for Thermal Module of SDTwin.

Document purpose: Define repeatable functional and numerical verification of the thermal module against the SDTwin Thermal Module Design Report and record the execution results of all cases.

1.2 Software and Document Overview

This report is intended for thermal, power and computing developers, system testers, and personnel responsible for asset configuration and model integration. The object under test is the lumped thermal model specified by the design report: the six temperature nodes solar array, computing node, cold plate, battery, power equipment and common radiator, the five heat-transfer connections SR, JC, CR, BR and DR, the four common data objects of Section 5.1, and the five functions of Chapter 5:

- `assemble_thermal_parameters`: assembles thermal capacitances, thermal resistances and surface parameters from the assets and the scene.
- `prepare_surface_environment`: prepares the irradiation inputs of every surface at one instant.
- `calculate_surface_heat`: calculates the environmental absorption and surface emission of the solar array and the radiator by Equation T4.
- `calculate_heat_flows`: calculates the signed heat flows of the five connections by Equation T2.
- `thermal_derivative`: calculates the temperature derivatives of the six components by Equation T3.

The test cases follow Equations T1 to T5 in Chapter 4, the data objects and functions in Chapter 5, the run procedure in Chapter 7, the numerical settings in Chapter 8 and the asset assembly requirements in Chapter 9 of the design report; the Design Basis row of each case names the sections, equations and functions it verifies. The thermal module is implemented as the thermal package of the design report, with the names, fields and call order of the four data objects and five functions unchanged. The end-to-end cases also need the coupled integration procedure, the Earth albedo and infrared calculation, the scene assembly and Power results; these test support programs are in the `sdtwin_sim` directory. The Power program used for testing follows the SDTwin Power Module Design Report, and its battery parameters are illustrative test values, not a selected device.

All 16 cases were executed on 6 October 2026. The actual result and conclusion of each case are recorded in Chapter 3 and summarized in Chapter 7.

1.3 Abbreviations and Conventions

Term	Definition
Design report	SDTwin Thermal Software Module Design Report, issue of 2 October 2026
Design basis	Design report sections, equations, and functions that a test case verifies
Lumped model	Lumped thermal RC model of Chapter 4 of the design report, in which one temperature represents each component
Finite-element model	COMSOL finite-element thermal model, comprising the complete International Space Station model and the whole-satellite model of a single satellite, which calculates the temperature distributions over the surfaces and within the interiors of parts on a mesh
ISS	International Space Station
EATCS	External Active Thermal Control System of the ISS, which carries equipment heat to the radiators through liquid ammonia loops
MBSU, DDCU, IEA	Main Bus Switching Unit, DC-to-DC Converter Unit, and Integrated Equipment Assembly of the ISS, all three mounted on liquid ammonia cold plates
GPU	Graphics processing unit, used as the computing chip in this project
Third orbit	Last orbital period of the ISS finite-element calculation; the statistics of FE-001 to FE-003 are taken over this orbit
Component symbols	S , J , C , B , D , and R denote the solar array, computing node, cold plate, battery, power equipment, and common radiator respectively, as in the design report
sdtwin_sim	Directory of test support programs, containing the coupled integration procedure, the Earth albedo and infrared calculation, scene assembly, and the test power program
Test power program	Power functions written according to the Power module design report, which supply the four power ports, battery state derivatives, and event decisions for the end-to-end test cases; the battery parameters are illustrative test values

Temperatures inside the module are in K. The finite element comparisons give temperatures in °C and temperature differences in K; minimum, mean and maximum temperatures are time-weighted over the exact orbit period.

2 Test Content

The verification scope covers the four requirements TH-01 to TH-04 in Chapter 11 of the design report with 16 cases organized into six groups that follow the design report chapters. Each case is traced to one function, data object or system-level comparison and to one final summary record.

- Common Data and Parameter Assembly: DM-001, PA-001.

- Surface Environment and Radiation: EN-001, EN-002, EN-003.
- Component Heat Transfer and Temperature Derivatives: HT-001, HT-002, HT-003.
- Electrical/Thermal Coupling: EC-001.
- Finite-Element Benchmarks: FE-001, FE-002, FE-003, FE-004.
- Numerical Calculation and End-to-End Operation: NI-001, NI-002, NI-003.

Table 1. Functional Requirement and Test-Case Traceability

No.	Case ID	Function	Test Name	Requirement	Status
1	DM-001	ThermalParameters, ThermalState, ThermalInputs, ThermalEvaluation	Common Data Objects	TH-01, TH-02, TH-04	Pass
2	PA-001	assemble_thermal_parameters	Thermal Capacitance and Resistance Assembly	TH-04	Pass
3	EN-001	prepare_surface_environment	Surface Environment Preparation	TH-03	Pass
4	EN-002	calculate_surface_heat	Surface Absorption and Emission	TH-03	Pass
5	EN-003	earth_flux environment input	Earth Albedo and Infrared Inputs	TH-03	Fail
6	HT-001	calculate_heat_flows	Component Heat- Transfer Calculation	TH-01	Pass
7	HT-002	thermal_derivative	Temperature Derivatives and Overall Heat Balance	TH-01	Pass
8	HT-003	thermal_derivative and the numerical solver	Comparison with Analytical Solutions	TH-01	Pass
9	EC-001	ThermalInputs and thermal_derivative	Electrical/Thermal Coupling	TH-02	Pass
10	FE-001	Solar array S	Solar Array Node Finite-Element Benchmark	TH-01, TH-03	Fail
11	FE-002	Common radiator R	Radiator Node Finite-Element Benchmark	TH-01, TH-03	Pass
12	FE-003	Computing node J and cold plate C	Computing Node and Cold Plate Finite- Element Benchmark	TH-01, TH-04	Pass
13	FE-004	Computing node, cold plate, and radiator chain	Whole-Satellite Finite-Element Chain Benchmark	TH-01, TH-04	Fail

No.	Case ID	Function	Test Name	Requirement	Status
14	NI-001	Numerical solver settings	Numerical Calculation and Eclipse Boundaries	TH-01	Pass
15	NI-002	Joint operation and result archive	End-to-End Operation and Result Archive	TH-01, TH-02	Pass
16	NI-003	SimReady assets and scene configuration	Asset and Scene Assembly	TH-04	Pass

3 Detailed Test Projects

The following records use the same execution structure as the reference outline; the Design Basis row names the design report content each case verifies. The numerical thresholds are default verification gates and shall be tightened when an approved module requirement defines a stricter value. Cases with finite element comparisons give the comparison tables and curves after the case table.

3.1 Common Data and Parameter Assembly

3.1.1 Test Project 1: Common Data Objects

This project verifies the four common data objects and the package exports of the SDTwin thermal module, specified in Section 5.1 and Section 3.2 of the design report respectively.

Table 2. DM-001 - Common Data Objects Test Case

Case ID	DM-001	Case Name	Common Data Objects
Test Objective	Verify the fields, array dimensions, array order, and construction checks of ThermalParameters, ThermalState, ThermalInputs, and ThermalEvaluation, and check the public exports of the thermal package.		
Design Basis	Section 3.2; Section 5.1, Table 6; Section 6.1; Appendix A		
Preconditions	The four data objects are implemented according to Table 6 of the design report, and the thermal package is organized according to Section 3.2.		
Input Data	- Nominal construction: C_J_K holds six positive values, R_K_W holds five positive values, temperature_K holds six positive values, and the environment and four powers of ThermalInputs share a timestamp. - Q_B_W set to a positive value and, separately, to a negative value. - Invalid construction: mismatched array dimensions, nonfinite values, nonpositive thermal capacitance or resistance, and mismatched run identifiers or times.		

Procedure	1. Construct the four objects from the nominal inputs and read all fields. 2. Check that the temperature order is S, J, C, B, D, R, the path order is SR, JC, CR, BR, DR, and the environment and radiation array order is S, R. 3. Check that Q_B_W retains its sign. 4. Inject the invalid constructions one at a time and record the error messages. 5. Attempt to modify ThermalParameters after the run starts, and check that trial states and accepted states use separate records. 6. Check that <code>__init__.py</code> exports the four data types and five functions listed in Appendix A.		
Expected Results	• Fields, dimensions, and order conform to Table 6. • Invalid constructions are rejected, and each error message names the specific field. • The parameter object is read-only after the run starts, and trial states do not overwrite accepted states. • The public exports match Appendix A.		
Acceptance Criteria	All fields, dimensions, and orders shall match Table 6. All invalid constructions shall produce deterministic errors. The read-only, state-separation, and public-export checks shall all pass.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	The four records were checked for fields, array order, units, immutability, invalid construction and trial-state handling in coupled runs. Battery heat inputs of 10 W and -10 W give 0.004 K/s and -0.016 K/s; the latter is independently calculated from T3. After correcting package exports, the 13 public names comprise the nine names in Appendix A and four error or warning classes. SurfaceRecord and ordering constants are imported from thermal.types. The evidence also records rejected-step, event-boundary and temperature-ownership checks. Test program tests/test_dm_001_data_objects.py recorded 217 checks, 217 passed. Evidence file: tests/results/DM-001.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.1.2 Test Project 2: Thermal Capacitance and Resistance Assembly

This project verifies the function of the SDTwin thermal module that assembles component thermal capacitances and connection thermal resistances according to Equation T5.

Table 3. PA-001 - Thermal Capacitance and Resistance Assembly Test Case

Case ID	PA-001	Case Name	Thermal Capacitance and Resistance Assembly
Test Objective	Verify that assemble_thermal_parameters assembles thermal capacitances, thermal resistances, and surface parameters from asset materials and connections according to T5, and check override precedence, duplicate material assignment, equivalent total resistance, and provenance records.		
Design Basis	Section 4.5, Equation T5; Section 5.2; Section 9.4		

Preconditions	• components contains the instance-to-node assignments, material masses, specific heats, and surface records. • connections gives the length, thermal conductivity, cross-sectional area, and contact resistance of SR, JC, CR, BR, and DR, or gives an equivalent total resistance confirmed by test. • A hand-calculation table prepared from the same inputs has been reviewed.		
Input Data	• Example in Section 4.5 of the design report: a material with a mass of 1 kg and a specific heat of $1000 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$. • MBSU box from the ISS model parameter table: $0.94 \text{ m}\times 0.84 \text{ m}\times 0.51 \text{ m}$, density 300 kg/m^3, specific heat $900 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$; radiator panel areal density 8 kg/m^2, specific heat $900 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$. • A set of overrides that changes the material mass of one component and the contact resistance of one connection. • Invalid inputs: zero or negative mass, specific heat, length, thermal conductivity, or cross-sectional area; negative contact resistance; one material portion assigned to two components; duplicate surfaces; undefined connections; a contact resistance given again for an equivalent total resistance that already includes it.		
Procedure	1. Call assemble_thermal_parameters, read C_J_K and R_K_W, and compare them item by item with the hand-calculation table. 2. Reassemble with the overrides added and check that the overrides take precedence over the asset parameter set. 3. Check that heat-pipe or liquid-cooling equivalent paths read the total resistance directly and do not add an already included contact resistance again. 4. Inject the invalid inputs one at a time and record the error messages. 5. Check that ThermalParameters records the resolved values, units, asset revisions, and provenance, and check that the function does not create or update operating temperatures.		
Expected Results	• Thermal capacitances and resistances agree with the hand calculation, and the overrides take effect. • Each invalid input identifies the offending record, and no default device values are produced. • The function produces no operating temperatures.		
Acceptance Criteria	The relative error of thermal capacitances and resistances shall not exceed 1×10^{-12}. The thermal capacitance shall be 1000 J/K for the Section 4.5 example and 108.7 kJ/K for the MBSU box, and the radiator panel thermal capacitance per unit area shall be $7.2 \text{ kJ}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$. All invalid inputs shall produce deterministic errors.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	Six capacitances and five resistances were assembled from material mass and specific heat, conduction geometry and contact resistance, with explicit scene overrides. The largest relative error of baseline and overridden values against independent calculations is 1.00×10^{-16}. The design example gives 1000 J/K, the MBSU gives 108.7 kJ/K, and the radiator areal capacity is $7.2 \text{ kJ}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$. Rejection checks cover double-counted contact resistance, missing and non-finite parameters and invalid surfaces. Provenance records parameter sources and overrides. Test program tests/test_pa_001_t5_assembly.py recorded 114 checks, 114 passed. Evidence file: tests/results/PA-001.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.2 Surface Environment and Radiation

3.2.1 Test Project 3: Surface Environment Preparation

This project verifies the function of the SDTwin thermal module that prepares synchronized surface environment inputs.

Table 4. EN-001 - Surface Environment Preparation Test Case

Case ID	EN-001	Case Name	Surface Environment Preparation
Test Objective	Verify that prepare_surface_environment obtains cos_incidence for each surface from synchronized orbit, attitude, and surface records, and arranges the environment inputs in asset surface order.		
Design Basis	Section 5.3; Section 6.1		
Preconditions	- orbit_input contains run_id, time_s, epoch, position_m, sun_position_m, frame, quaternion_xyzw, and G_W_m2, and the satellite position and the Earth-to-Sun vector have been converted to GCRS. - earth_flux carries the same run identifier and time as orbit_input. - An independent vector and quaternion calculation program is available.		
Input Data	- Constructed attitudes: the identity quaternion, a 90° rotation about each of the three axes, and arbitrary combined rotations. - Surface normals: the six body face directions and one arbitrary direction. - Orbit output over one full revolution of a circular orbit at 400 km altitude, including the eclipse phase; G is given by sun_intensity with include_eclipse true. - Invalid inputs: an unnormalized quaternion or normal, a zero difference between the satellite position and the Sun vector, an unconverted TEME position, an unsupported frame, mismatched surface identifiers, and a missing earth_flux record.		
Procedure	- Subtract the satellite position from the Earth-to-Sun vector and normalize the result to obtain the satellite-to-Sun direction; calculate cos_incidence for each constructed attitude and compare it with the independent calculation. - Calculate one full orbit and check that the returned dictionary contains run_id, time_s, surface_ids, G_W_m2, cos_incidence, albedo_W_m2, and infrared_W_m2, and that the surface arrays have equal lengths and follow asset surface order. - Check that G_W_m2 is identical to the Orbit output at every point and is not multiplied by an eclipse factor again. - Inject the invalid inputs one at a time and record how each is handled.		
Expected Results	- cos_incidence agrees with the independent calculation, and surfaces facing away from the Sun receive negative values. - G is zero in eclipse and is supplied only by Orbit. - The function does not change temperatures, and invalid inputs are rejected.		
Acceptance Criteria	The cos_incidence error shall not exceed 1×10^{-12} . Over the full orbit, G shall be identical to the Orbit output at every point. All invalid inputs shall produce deterministic errors.		
Case Designer		Executor / Date	Automated test, 2026-10-06

Actual Results	The surface environment was evaluated at 5555 instants on a 400 km circular orbit using fixed rotations, random quaternions, attitude dynamics and Earth-pointing attitudes. The largest incidence-cosine error against independent rotation calculations is 7.77×10^{-16} against 1×10^{-12}. G is bitwise identical to Orbit output, with no second eclipse multiplier. The 51 invalid-input checks cover quaternions, normals, frames, surface order, timing and missing environment records. Test program tests/test_en_001_surface_environment.py recorded 88 checks, 88 passed. Evidence file: tests/results/EN-001.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.2.2 Test Project 4: Surface Absorption and Emission

This project verifies the function of the SDTwin thermal module that calculates the heat absorbed by exposed surfaces and their surface emission according to Equation T4.

Table 5. EN-002 - Surface Absorption and Emission Test Case

Case ID	EN-002	Case Name	Surface Absorption and Emission
Test Objective	Verify that calculate_surface_heat calculates, for each surface according to T4, the direct solar, albedo, and infrared absorption and the surface emission, sums them by S and R into Q_{env_W} and Q_{emit_W} , and returns absorbed_solar_S_W.		
Design Basis	Section 4.4, Equation T4; Section 5.4		
Preconditions	- Surface areas, absorptivities, and emissivities are assembled. - EN-001 has passed.		
Input Data	- Example in Section 4.4 of the design report: area 2 m^2, direct irradiance 1000 W/m^2, absorptivity 0.8. - Three orientations: facing the Sun, edge-on to the Sun, and facing away from the Sun; two environments: albedo only and infrared only. - Component temperatures of 200 K, 300 K, and 400 K. - Invalid inputs: zero or negative temperature, negative irradiance, mismatched run identifier or time, and a temperature outside the applicable range of the material parameters.		
Procedure	- Calculate Q_{env_W}, Q_{emit_W}, and absorbed_solar_S_W for each example and compare them with the hand calculation. - Check that direct irradiance is G multiplied by the nonnegative incidence cosine, that direct and albedo irradiance are multiplied by absorptivity and infrared by emissivity, and that the area is not projected again. - Check that surface emission is calculated from the fourth power of the absolute temperature of the owning component. - Check that the front and back of the solar array are calculated separately and then summed, and that absorbed_solar_S_W includes only the direct solar power absorbed by the solar array. - Inject the invalid inputs one at a time.		

Expected Results	• The absorbed power in the Section 4.4 example is 1600 W, and direct absorption is zero when the surface faces away from the Sun. • When a temperature lies outside the applicable material range, the limitation is reported. • The function does not modify the state.		
Acceptance Criteria	The relative error against the hand calculation shall not exceed 1×10^{-12} . The ratio of the emitted power at 400 K to that at 200 K shall be 16. All invalid inputs shall produce deterministic outcomes.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	T4 was calculated independently for combinations of orientation, optical properties, sunlight and Earth flux. The largest relative error of absorbed and emitted power is 1.93×10^{-16} against 1×10^{-12}. The Section 4.4 example gives 1600 W. Raising temperature from 200 K to 400 K gives an emission ratio of 16, as required by the fourth-power law. Internal nodes do not receive additional surface-radiation terms, and out-of-range temperatures produce warnings. Test program tests/test_en_002_surface_heat.py recorded 62 checks, 62 passed. Evidence file: tests/results/EN-002.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.2.3 Test Project 5: Earth Albedo and Infrared Inputs

This project verifies the Earth albedo and Earth infrared inputs supplied to Equation T4 of the SDTwin thermal module. Section 5.3 of the design report specifies that these two inputs come from additional environmental data and cannot be inferred from the solar irradiance of Orbit; this project checks that data source.

Table 6. EN-003 - Earth Albedo and Infrared Inputs Test Case

Case ID	EN-003	Case Name	Earth Albedo and Infrared Inputs
Test Objective	Verify the albedo_W_m2 and infrared_W_m2 values that earth_flux supplies for each surface_id, and confirm that the Earth view factor and the albedo calculation of each surface are correct.		
Design Basis	Section 4.4, Table 5; Section 5.3; Section 9.4		
Preconditions	• earth_flux is connected according to Section 5.3 and carries the same run identifier and time as orbit_input.		
Input Data	• Circular orbit at 400 km altitude, β angle 0° and 75°. • Albedo and Earth infrared at the ISS design case values: cold case 0.20 and 206 W/m², mean case 0.31 and 241 W/m², hot case 0.40 and 286 W/m². • Surface orientations: facing Earth, facing away from Earth, perpendicular to the local vertical, and tracking the Sun.		

Procedure	1. Compare the view factor of an Earth-facing flat plate with the analytical value, which equals the square of the ratio of the Earth radius to the orbit radius. 2. Compare the other orientations with numerical integration over the visible Earth surface. 3. Export the albedo and infrared heat loads of the solar array wings and radiators from the solved ISS finite-element model and compare them with the earth_flux results. 4. Delete one albedo or infrared record and check that the system identifies the unconfigured field according to Section 9.4 and does not treat the missing value as zero.		
Expected Results	• The view factor of the Earth-facing plate agrees with the analytical value. • The albedo and infrared irradiance for each orientation agree with the numerical integration. • When a record is missing, the unconfigured field is reported.		
Acceptance Criteria	The view factor of the Earth-facing plate at 400 km shall be 0.8855, with an error not exceeding 0.1%. For the other orientations, differences from the numerical integration shall not exceed 1%. Differences from the finite-element heat loads shall not exceed 3%. All missing records shall be reported.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	earth_flux was checked against the analytic nadir-plate view factor, independent geocentric quadrature and Monte Carlo integration. The relative criterion remains 1%; only numerical zero uses an absolute tolerance of 1×10^{-12} W/m². The original FE comparison retains the 2×6 planet discretisation and applies the 3% criterion to simultaneous loads, orbit means and peaks. Radiator comparisons use the outer faces without obstruction by the other wing; inner faces are diagnostic. Original-reference exceedances remain in the following tables. A further 3 cases compare 2×6, 4×12 and 8×24 planet discretisations using mean loads at fixed sample instants. Refinement verifies the reference; it neither replaces the original acceptance nor establishes convergence of the full temperature simulation. Test program tests/test_en_003_earth_flux.py recorded 89 checks, 72 passed, 17 failed. Evidence file: tests/results/EN-003.json.		
Recorded By		Test Conclusion	[] Pass [X] Fail [] Not Executed
Anomalies	The case does not meet all recorded requirements. The measured exceedances and their evidence limits are stated above and in the comparison tables. Failed check count: 17. See tests/results/EN-003.json for each expected value, actual value and verdict.		

Table 7 gives the step 3 comparison of thermal loads. The FE values come from the solved models, whose planet is discretised into 2 rings of 6 points. Differences are earth_flux minus FE divided by FE; peak differences compare the two peaks. Pointwise maxima compare nonzero FE values at the same instants. Tiny reference values can produce very large relative errors, so absolute weak-flux values must also be examined.

Table 7. EN-003 Earth albedo and infrared thermal loads, solved models

Case	Thermal load	earth_flux mean kW	FE mean kW	Mean diff %	Peak diff %	Pointwise max %
Design cold	Solar array albedo	87.03	88.40	−1.6	1.8	100.0
Design cold	Solar array infrared	303.31	315.25	−3.8	1.8	13.4
Design cold	Radiator albedo	1.60	1.58	1.2	1.2	79.3
Design cold	Radiator infrared	28.85	28.47	1.3	1.9	1.9
Mean environment	Solar array albedo	140.00	142.20	−1.6	1.8	100.0
Mean environment	Solar array infrared	354.85	368.81	−3.8	1.8	13.4
Mean environment	Radiator albedo	3.43	3.39	1.2	1.2	79.3
Mean environment	Radiator infrared	33.75	33.31	1.3	1.9	1.9
Design hot	Solar array albedo	44.33	52.61	−15.7	−10.1	7.81×10^{11}
Design hot	Solar array infrared	344.92	379.08	−9.0	−8.5	10.1
Design hot	Radiator albedo	2.32	2.37	−2.2	−0.7	72.2
Design hot	Radiator infrared	49.44	48.50	1.9	1.9	2.0

The largest pointwise relative difference occurs for solar array albedo in the design hot case at 13320 s. The module gives 407.83 W and FE gives 5.220×10^{-8} W, an absolute difference of 407.83 W. The near-zero reference produces a large relative difference. Pointwise values, time-weighted means and peaks retain separate 3% checks.

To verify the reference, fe_refine_ladder.py re-solved only the orbital thermal loads of the same models with successively finer planet discretisations at 16 instants spaced 360 s from 10920 to 16320 s. These are arithmetic sample means, not time-weighted means over an exact orbit. Table 8 gives the mean FE loads of each level at these instants, the earth_flux mean at

the same instants, and the difference of earth_flux from the finest level and from a conditional extrapolation. Extrapolation is shown only when successive changes have the same sign and shrink; otherwise it is N/A. Extra eclipse-boundary samples saved by COMSOL remain in the raw exports but are excluded from the common sample means. Three levels alone do not establish asymptotic convergence.

Table 8. EN-003 thermal loads with refined FE planet discretisation

Case	Thermal load	FE 2×6 kW	FE 4×12 kW	FE 8×24 kW	Module kW	vs finest %	vs extrap. %
Design cold	Solar array albedo	95.27	95.58	94.91	94.30	−0.6	N/A
Design cold	Solar array infrared	316.82	310.91	307.55	305.63	−0.6	0.8
Design cold	Radiator albedo	1.69	1.75	1.72	1.71	−0.4	N/A
Design cold	Radiator infrared	28.30	29.10	28.80	28.67	−0.4	N/A
Mean environment	Solar array albedo	153.25	153.76	152.67	151.70	−0.6	N/A
Mean environment	Solar array infrared	370.65	363.73	359.81	357.56	−0.6	0.8
Mean environment	Radiator albedo	3.63	3.76	3.69	3.68	−0.4	N/A
Mean environment	Radiator infrared	33.11	34.05	33.69	33.54	−0.4	N/A
Design hot	Solar array albedo	55.47	50.16	48.08	47.06	−2.1	0.7
Design hot	Solar array infrared	379.14	354.46	346.73	345.13	−0.5	0.6
Design hot	Radiator albedo	2.53	2.53	2.50	2.48	−0.9	N/A
Design hot	Radiator infrared	48.48	49.64	49.60	49.41	−0.4	N/A

Across these sample means, the largest module-to-finest relative differences are Design cold 0.6%, Mean environment 0.6%, Design hot 2.1%. Refining the planet from 2×6 to 8×24 changes the FE mean by up to 13.3%, for solar array albedo in the design hot case. This demonstrates sensitivity of the original reference to planet discretisation. This comparison does not verify full-orbit pointwise or peak errors and does not re-solve the temperature field, so it does not replace the original acceptance checks.

For weak albedo, the independent geocentric reference separates ring and azimuth integration, splits at illumination and visibility boundaries and uses adaptive integration to check uniform grids. The following mean-environment examples select the largest absolute

and relative exceedances. The default earth_flux grid therefore does not guarantee a uniform 1% relative accuracy at weak flux.

Table 9. EN-003 representative weak-albedo errors

β °	Normal	Orbit angle °	Module W/m ²	Reference W/m ²	Abs diff W/m ²	Rel diff %
0	Sun facing	101.25	0.24661828	0.24979921	0.00318093	1.27
0	Nadir	108.75	0	3.750×10^{-5}	3.750×10^{-5}	100.00
75	Sun facing	221.25	0.24999912	0.25319894	0.00319982	1.26
75	Cross track +	108.75	0	9.046×10^{-10}	9.046×10^{-10}	100.00

3.3 Component Heat Transfer and Temperature Derivatives

3.3.1 Test Project 6: Component Heat-Transfer Calculation

This project verifies the function of the SDTwin thermal module that calculates connection heat flows according to Equation T2.

Table 10. HT-001 - Component Heat-Transfer Calculation Test Case

Case ID	HT-001	Case Name	Component Heat-Transfer Calculation
Test Objective	Verify that calculate_heat_flows evaluates T2 in the fixed order SR, JC, CR, BR, DR and returns signed heat flows.		
Design Basis	Section 4.2, Equation T2; Section 5.5		
Preconditions	PA-001 has passed, and the thermal resistances of the five connections are assembled.		
Input Data	- Battery example in Section 4.3 of the design report: T_B is 303 K, T_R is 300 K, and R_{BR} is 0.5 K/W. - Cases with the two endpoint temperatures swapped and with equal endpoint temperatures. - Invalid inputs: a missing node, mismatched array order, zero or negative resistance, and nonfinite inputs.		
Procedure	- Calculate the five connection heat flows, compare them with the hand calculation, and check that the returned dictionary contains run_id, time_s, and q_W. - Swap the two endpoint temperatures and check that the heat flow changes sign while its magnitude is unchanged. - Check that a positive value means transfer from the first node in the path name to the second. - Inject the invalid inputs one at a time.		
Expected Results	- The example heat flow q_{BR} is 6 W, -6 W after the swap, and zero at equal temperatures. - Each path is evaluated once, and the function performs no state update. - Invalid inputs produce errors, and neither zero resistance nor ad hoc clipping of the heat flow stands in for the actual path.		

Acceptance Criteria	The relative error against the hand calculation shall not exceed 1×10^{-12} . All invalid inputs shall produce deterministic errors.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	All five heat paths were compared with independent T2 calculations, giving a largest relative error of 9.67×10^{-17}. The battery example gives 6 W, changes to -6 W when the end temperatures are exchanged, and gives zero for equal temperatures. Checks also record temperature and resistance access order, evaluations per path, extreme valid resistances and rejection of invalid inputs. Test program tests/test_ht_001_heat_flows.py recorded 91 checks, 91 passed. Evidence file: tests/results/HT-001.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.3.2 Test Project 7: Temperature Derivatives and Overall Heat Balance

This project verifies the temperature derivatives of the SDTwin thermal module, formed according to Equation T3, and checks the overall heat balance of Equation T1.

Table 11. HT-002 - Temperature Derivatives and Overall Heat Balance Test Case

Case ID	HT-002	Case Name	Temperature Derivatives and Overall Heat Balance
Test Objective	Verify that thermal_derivative returns, in one call according to T3, the temperature derivatives of S , J , C , B , D , and R and the ThermalEvaluation, and that the result satisfies T1.		
Design Basis	Section 4.1, Equation T1; Section 4.3, Equation T3; Section 5.6		
Preconditions	• EN-002 and HT-001 have passed.		
Input Data	• Battery example in Section 4.3 of the design report: C_B is 1000 J/K, T_B is 303 K, T_R is 300 K, R_{BR} is 0.5 K/W, and Q_B is 10 W. • A total of 1000 randomly generated sets of temperatures, four powers, and environment inputs.		
Procedure	1. Calculate the heating rate of the battery example. 2. For the random inputs, calculate the sum of the products of component thermal capacitance and temperature derivative, and compare it with the right-hand side of T1. 3. Check that each heat flow is subtracted at the outgoing node and added at the incoming node. 4. Check that ThermalEvaluation contains $dT_dt_K_s$, q_W, Q_env_W, Q_emit_W, T_B_K, and T_J_K, and that T_B_K and T_J_K copy the corresponding input temperatures and are not new integrated states.		
Expected Results	• The heating rate of the battery example is 0.004 K/s. • The summed rate of heat storage of all components equals the right-hand side of T1.		
Acceptance Criteria	The error in the heating rate of the example shall not exceed 1×10^{-12} K/s. The ratio of the overall heat-balance residual to the total input power shall not exceed 1×10^{-12} .		
Case Designer		Executor / Date	Automated test, 2026-10-06

Actual Results	The design battery example gives a derivative of 0.004 K/s. With a fixed random seed, 1000 input sets were checked against node balances T3 and total balance T1. The largest T1 residual divided by input heat power is 8.75×10^{-16} against 1×10^{-12}. All five internal flows cancel in the total balance. Returned T_B_K and T_J_K are temperatures from the input state. Test program tests/test_ht_002_derivative_balance.py recorded 21 checks, 21 passed. Evidence file: tests/results/HT-002.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.3.3 Test Project 8: Comparison with Analytical Solutions

This project verifies the temperature integration results of the SDTwin thermal module using examples that have analytical solutions.

Table 12. HT-003 - Comparison with Analytical Solutions Test Case

Case ID	HT-003	Case Name	Comparison with Analytical Solutions
Test Objective	Verify the temperature evolution over time obtained by combining thermal_derivative with the numerical solver of Chapter 8.		
Design Basis	Section 4.3, Equation T3; Chapter 8, Table 8		
Preconditions	- HT-002 has passed. - RK45 is selected as the initial integration method according to Chapter 8, and the tolerances and maximum step are configured.		
Input Data	- Single-node step: the cold-plate temperature is held constant, P_{load} of the computing node steps from zero to 300 W, C_f is 500 J/K, and R_{JC} is 0.1 K/W. - Series steady state: the computing node connects through the cold plate to a radiator held at constant temperature, and P_{load} is 300 W. - Radiative equilibrium: the radiator has only a fixed input power and its own surface emission.		
Procedure	- Integrate the single-node step and compare it with the exponential solution. - Integrate the series example to steady state; the temperature difference between the computing node and the radiator should equal the power multiplied by the sum of R_{JC} and R_{CR}. - Integrate the radiative-equilibrium example to steady state; the fourth power of the steady-state temperature should equal the input power divided by the product of emissivity, the Stefan-Boltzmann constant, and the radiating area.		
Expected Results	The numerical solutions of the three examples agree with the analytical solutions.		
Acceptance Criteria	The step-response error shall not exceed 0.001 K over the whole run. The errors of the two steady-state examples shall not exceed 0.001 K.		
Case Designer		Executor / Date	Automated test, 2026-10-06

Actual Results	Three analytic cases were run through the coupled solver: computing-node step response, series-resistance equilibrium and radiator radiation balance. The largest step-response error is 1.79×10^{-6} K, the series steady-temperature-drop error is 3.75×10^{-8} K, and the radiation equilibrium error is 4.09×10^{-12} K. Steady errors are compared with 0.001 K. Fixed-temperature boundaries use declared large capacitances and isolation resistances, with boundary drift checked separately. Test program tests/test_ht_003_analytic.py recorded 27 checks, 27 passed. Evidence file: tests/results/HT-003.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.4 Electrical/Thermal Coupling

3.4.1 Test Project 9: Electrical/Thermal Coupling

This project verifies the function of the SDTwin thermal module that uses the four power ports of Power.

Table 13. EC-001 - Electrical/Thermal Coupling Test Case

Case ID	EC-001	Case Name	Electrical/Thermal Coupling
Test Objective	Verify that P_{pv_W} , P_{load_W} , Q_{B_W} , and Q_{D_W} enter the corresponding components according to T3, and check the sign constraints, the upper limit on photovoltaic output, the validity of the Power result, and the single ownership of battery temperature.		
Design Basis	Section 4.1; Section 4.3; Section 5.6; Section 6.1		
Preconditions	- HT-002 has passed. - solve_power_allocation of Power is available, or inputs constructed from the PowerResult fields are used.		
Input Data	P_{pv} takes values between zero and the current absorbed_solar_S_W; P_{load} takes values between zero and the rated power; Q_B takes positive and negative values; Q_D takes positive values. - Invalid inputs: a negative P_{pv_W}, P_{load_W}, or Q_{D_W}; P_{pv} greater than absorbed_solar_S_W; valid false; event_required true.		
Procedure	- Change one power at a time and check that only the derivative of the corresponding component changes, by an amount equal to the power change divided by the thermal capacitance of that component: P_{pv} affects only S, P_{load} only J, Q_B only B, and Q_D only D. - Set Q_B to a negative value and check that the battery temperature derivative decreases accordingly. - Reduce P_{pv} to represent the controller curtailing generation, and check that the energy not exported remains in the energy balance of the solar array. - Inject the invalid inputs one at a time. - Run one orbit jointly with Power and confirm that the battery temperature is updated only in the thermal module and is returned to Power as T_{B_K}.		

Expected Results	• Each power enters the correct component, and its sign is retained. • An invalid Power result does not enter the thermal derivative; the thermal module neither repairs invalid electrical powers nor decides load restart. • Battery temperature is updated only once.		
Acceptance Criteria	The relative error of the derivative changes against the hand calculation shall not exceed 1×10^{-12} . All invalid inputs shall produce deterministic outcomes. During the joint run, the battery temperature shall be updated only in the thermal module.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	Each Power port was varied independently. The largest relative error of derivative changes against T3 is 6.05×10^{-16}. Tests cover negative battery heat, PV curtailment, PV output exceeding absorbed sunlight, and handling of invalid, supply_shortfall and device_boundary results. The coupled run records 1700 Power calls, each using battery temperature from the same thermal evaluation. The Power validation copy agrees with the thermal initial temperature; Power does not integrate a separate battery temperature. Test program tests/test_ec_001_power_ports.py recorded 83 checks, 83 passed. Evidence file: tests/results/EC-001.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.5 Finite-Element Benchmarks

This group compares thermal module results with finite element results. The module represents each component by one temperature, so the comparison uses the mean temperature of the same part in the finite element model. The finite element result files are listed in Chapter 6.

3.5.1 Test Project 10: Solar Array Node Finite-Element Benchmark

This project verifies the solar array node of the SDTwin thermal module against finite-element results for the US solar array wings of the ISS.

Table 14. FE-001 - Solar Array Node Finite-Element Benchmark Test Case

Case ID	FE-001	Case Name	Solar Array Node Finite-Element Benchmark
Test Objective	Verify the solar array temperature calculated by the first line of T3 and by T4, and check environmental absorption, the subtraction of actual electrical output, and surface emission from the front and back faces.		
Design Basis	Section 4.3, first line of Equation T3; Section 4.4, Equation T4; Section 4.5, Equation T5		
Preconditions	• The third-orbit time histories of the three finite-element cases and the model parameter table are available. • The attitude quaternion keeps the front normal of the solar array pointed at the Sun, matching the Sun-pointing law of the solar array wings in the finite-element model. • The finite-element model excludes conduction between the solar array wings and the radiators; the SR connection takes a very large resistance to represent no conduction, and this test configuration is recorded according to Section 9.4.		

Input Data	• Front absorptivity 0.72 and emissivity 0.82, back absorptivity 0.55 and emissivity 0.85, and thermal capacitance per unit area $1.6 \text{ kJ}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$, all taken from the finite-element model parameter table. • P_{pv} taken as the generation fraction 0.073 multiplied by the direct solar power received by the solar array, representing the actual electrical output supplied by Power. • Design cold case: β angle 0°, solar constant 1321 W/m^2, albedo 0.20, Earth infrared 206 W/m^2. • Mean environment case: β angle 0°, solar constant 1371 W/m^2, albedo 0.31, Earth infrared 241 W/m^2. • Design hot case: β angle 75°, solar constant 1423 W/m^2, albedo 0.40, Earth infrared 286 W/m^2. • Circular orbit at 400 km altitude with a period of 5553.6 s.		
Procedure	1. Generate orbit_input and earth_flux for each case. 2. Integrate with the numerical settings of Chapter 8 until successive orbits repeat, splitting the integration at eclipse boundaries, and take the last orbit. 3. Compare the minimum, mean, and maximum temperatures with the third orbit of the finite-element mean solar array wing temperature. 4. Recalculate once with albedo and infrared disabled, and record this simplification and the temperature difference according to Section 9.4.		
Expected Results	• The temperature curves of the three cases coincide with those of the finite-element model. • With albedo and infrared disabled, the temperatures are markedly lower.		
Acceptance Criteria	For all three cases, the differences between the minimum, mean, and maximum temperatures and the finite-element results shall not exceed 3 K.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	The lumped model uses ISS array area, areal heat capacity, front and back optical properties and three environmental cases. The largest absolute difference among nine third-orbit minimum, mean and maximum statistics is 1.69 K, meeting 3 K. However, simultaneous pointwise comparison at FE output instants gives maxima of 9.18 K, 12.09 K and 1.46 K for the cold, mean and hot cases. The cold and mean curves exceed the requirement, so the case fails. No time shift is allowed. Differences concentrate after eclipse transitions; current evidence does not attribute all of them to FE time stepping. Runs without albedo and infrared are recorded separately. Test program tests/test_fe_001_solar_array_fe.py recorded 71 checks, 68 passed, 3 failed. Evidence file: tests/results/FE-001.json.		
Recorded By		Test Conclusion	☐ Pass ☒ Fail ☐ Not Executed
Anomalies	The case does not meet all recorded requirements. The measured exceedances and their evidence limits are stated above and in the comparison tables. Failed check count: 3. See tests/results/FE-001.json for each expected value, actual value and verdict.		

Table 15 and Figure 1 compare the thermal module with the FE mean temperature of the solar array wings over the last orbit.

Table 15. FE-001 solar array temperatures over the last orbit

Case	FE min / mean / max °C	Thermal module min / mean / max °C	Max difference K
Design cold	−80.88 / 3.99 / 57.06	−82.35 / 3.37 / 57.68	1.47
Mean environment	−74.08 / 10.67 / 66.36	−75.78 / 9.72 / 67.17	1.69
Design hot	53.14 / 54.70 / 57.25	52.11 / 53.49 / 56.11	1.20

Table 16. FE-001 simultaneous pointwise temperature differences

Case	Pointwise max K	RMS K	Pointwise limit K
Design cold	9.18	2.89	3.00
Mean environment	12.09	3.51	3.00
Design hot	1.46	1.21	3.00

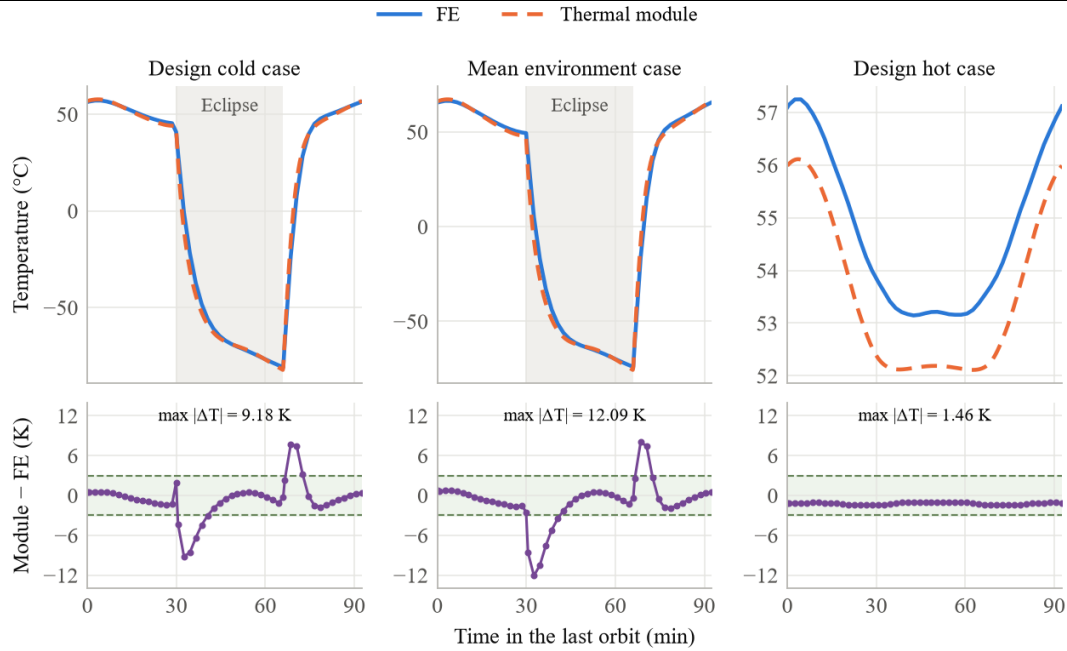


Figure 1. FE-001 solar array temperatures over the last orbit

The lower panels compare temperatures at FE output instants; the green band denotes ± 3 K. Close temperature statistics can coexist with local curve exceedances after eclipse transitions.

With Earth albedo and infrared switched off, the maximum temperature is 15.3 to 29.2 K lower and the minimum temperature of the cases with eclipse is 57.1 to 63.6 K lower, so the `earth_flux` input required by Section 5.3 of the design report is indispensable.

3.5.2 Test Project 11: Radiator Node Finite-Element Benchmark

This project verifies the radiator node of the SDTwin thermal module against finite-element results for the radiators of the ISS External Active Thermal Control System.

Table 17. FE-002 - Radiator Node Finite-Element Benchmark Test Case

Case ID	FE-002	Case Name	Radiator Node Finite-Element Benchmark
Test Objective	Verify the radiator temperature calculated by the sixth line of T3 and by T4, and check the branch heat inputs, environmental absorption, two-sided surface emission, and thermal capacitance.		
Design Basis	Section 4.3, sixth line of Equation T3; Section 4.4, Equation T4; Section 4.5, Equation T5; Chapter 10		
Preconditions	- The finite-element time histories of radiator heat input for each loop and the mean panel temperatures of the three radiator units are available. - The sum of q_{SR}, q_{CR}, q_{BR}, and q_{DR} is replaced by the finite-element radiator heat input. - The attitude quaternion follows the finite-element radiator pointing law: edge to the Sun in sunlight and face to the Earth in eclipse.		
Input Data	- Each loop: 24 panels, one-side area 220.7 m^2; C_R calculated by T5 from panel mass and specific heat, with the thermal capacitance of the liquid ammonia added, 1.61 MJ/K in total. - Z-93 white coating: absorptivity 0.15 in the cold case, 0.20 in the mean case, and 0.24 in the hot case; emissivity 0.90 to 0.91. - Design cold case: β angle 0°, solar constant 1321 W/m^2, albedo 0.20, Earth infrared 206 W/m^2. - Mean environment case: β angle 0°, solar constant 1371 W/m^2, albedo 0.31, Earth infrared 241 W/m^2. - Design hot case: β angle 75°, solar constant 1423 W/m^2, albedo 0.40, Earth infrared 286 W/m^2.		
Procedure	- Drive the radiator node with the finite-element heat-input time history, integrate three orbits from the finite-element initial temperature, and split the integration at eclipse boundaries. - Take the third orbit and compare it with the average of the mean panel temperatures of the three radiator units. - Record the difference between the coldest finite-element panel and the mean temperature at the same instant.		
Expected Results	- The mean temperature is close to the finite-element result. - Radiative obstruction between the two wings and panel temperature differences along the flow direction are possible residual sources. Section 4.4 and Chapter 10 of the design report state that this version neglects mutual obstruction between components, radiation exchange between components, and local temperature gradients.		
Acceptance Criteria	For both loops in all three cases, the third-orbit mean temperature difference shall not exceed 5 K.		
Case Designer		Executor / Date	Automated test, 2026-10-06

Actual Results	Node R uses FE radiator area, panel heat capacity and ammonia-loop capacity. FE loop heat input is prescribed as a boundary condition for separate A and B loop runs in three environments. The largest absolute third-orbit mean-temperature difference is 3.39 K against 5 K. Environmental absorption, emission and integration are checked independently. FE coldest-panel temperatures and panel spreads are recorded separately; passing a mean-temperature criterion does not validate local minima or hot spots. Obstruction and along-flow gradients are possible residual contributors, but their individual effects are not isolated here. Test program tests/test_fe_002_radiator_fe.py recorded 54 checks, 54 passed. Evidence file: tests/results/FE-002.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

Table 18 and Figure 2 compare the third-orbit mean temperature of the common radiator; the FE value is the average of the panel mean temperatures of the three radiator units.

Table 18. FE-002 common radiator third-orbit mean temperatures

Case	Loop	FE °C	Thermal module °C	Module minus FE K
Design cold	A	−54.85	−53.46	1.39
Design cold	B	−54.64	−51.25	3.39
Mean environment	A	−47.32	−46.61	0.71
Mean environment	B	−47.38	−44.57	2.81
Design hot	A	−36.25	−33.32	2.93
Design hot	B	−31.56	−31.69	−0.13

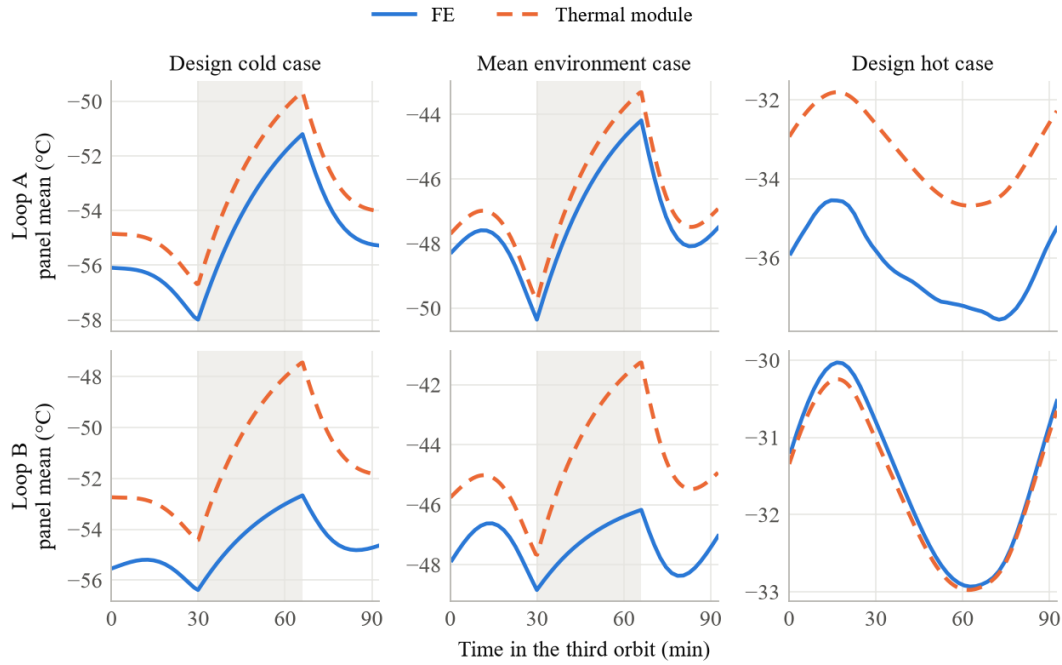


Figure 2. FE-002 common radiator temperatures over the third orbit

At the same instant the coldest FE panel is 18.8 to 20.6 K below the radiator mean. Chapter 10 of the design report states that the radiator uses one lumped temperature without local gradients, so freeze protection needs a separate coldest-point calculation. The FE model includes wing obstruction, which Section 4.4 excludes from this version; its separate temperature contribution has not been isolated.

3.5.3 Test Project 12: Computing Node and Cold Plate Finite-Element Benchmark

This project verifies the computing-node-to-cold-plate heat-transfer calculation of the SDTwin thermal module against finite-element results for ISS cold-plate equipment. Chapter 10 of the design report provides that finite elements may identify equivalent resistances for uncertain installation paths; on that basis, this project checks the composition of the JC resistance.

Table 19. FE-003 - Computing Node and Cold Plate Finite-Element Benchmark Test Case

Case ID	FE-003	Case Name	Computing Node and Cold Plate Finite-Element Benchmark
Test Objective	Verify the second line of T3 and the second line of T5, and confirm that the mean temperature of uniformly heated equipment can be represented by R_{JC} .		
Design Basis	Section 4.3, second and third lines of Equation T3; Section 4.5, Equation T5; Chapter 10		
Preconditions	- The finite-element part statistics and loop heat breakdown of the mean environment case are available. - The cold-plate thermal capacitance takes a very large value and starts at the coolant temperature, and the CR connection takes a very large resistance, so that the cold-plate temperature stays at the coolant temperature, matching the cold-plate boundary of the finite-element model.		

Input Data	• MBSU 495 W, cold-plate face 0.79 m²; DDCU 694 W, cold-plate face 0.56 m²; IEA 6000 W, cold-plate face 13.5 m². • Cold-plate heat-transfer coefficient 60 W·m⁻²·K⁻¹, equivalent equipment thermal conductivity 150 W·m⁻¹·K⁻¹, coolant 2.8 °C. • P_{load} apportions the FE loop cold-plate heat input among the devices. The roughly 5.4% remainder includes external radiation and storage, which current data cannot separate. The design-report computing node does not include an external surface-radiation path.		
Procedure	1. Calculate R_{JC} by the second line of T5: the contact part is the reciprocal of the product of the cold-plate heat-transfer coefficient and the cold-plate area; the conduction part uses one third of the equipment thickness as the conduction length, which is the equivalent length for the mean temperature when heat is generated uniformly within the equipment. 2. Integrate to steady state, where T_j equals the cold-plate temperature plus P_{load} multiplied by R_{JC}. 3. Compare with the finite-element third-orbit mean temperature.		
Expected Results	• The mean temperatures of the three equipment types agree with the finite-element results.		
Acceptance Criteria	The mean temperature difference shall not exceed 1 K.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	The computing-node to cold-plate path was compared for 14 ISS MBSU, DDCU and IEA units using mean-case coolant temperature as the cold-plate boundary. The largest absolute device mean-temperature difference is 0.25 K against 1 K. Large boundary capacitance, isolation resistance and sources are archived, and actual solver settings are compared with the configuration. Per-device cold-plate input is approximated using each loop share. The roughly 5.4% remainder includes both external radiation and storage; available data cannot separate them. Test program tests/test_fe_003_compute_cold_plate_fe.py recorded 63 checks, 63 passed. Evidence file: tests/results/FE-003.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

Table 20 gives the composition of R_{JC} and the mean temperatures by device class; temperatures are averages over the units of a class and the maximum difference is the largest absolute difference of a single unit.

Table 20. FE-003 computing node mean temperatures

Device	No.	P_{load} W	Cold plate area m ²	Contact K/W	Solid K/W	R_{JC} K/W	Module °C	FE °C	Max diff K
MBSU	4	468	0.7896	0.02111	0.00144	0.02254	13.34	13.54	0.23
DDCU	6	657	0.5609	0.02971	0.00353	0.03324	24.60	24.40	0.25
IEA	4	5836	13.5	0.00123	0.00026	0.00150	11.52	11.53	0.01

3.5.4 Test Project 13: Whole-Satellite Finite-Element Chain Benchmark

This project verifies the heat-transfer chain of the SDTwin thermal module in Figure 1, from the computing node through the cold plate to the radiator, against finite-element results for a single satellite.

Table 21. FE-004 - Whole-Satellite Finite-Element Chain Benchmark Test Case

Case ID	FE-004	Case Name	Whole-Satellite Finite-Element Chain Benchmark
Test Objective	Verify whether the three components J , C , and R and the two connections JC and CR can represent the temperature drop from the computing equipment to the radiator.		
Design Basis	Section 3.1, Figure 1; Section 4.3, Equation T3; Appendix B		
Preconditions	- The whole-satellite COMSOL results completed on 30 August 2026 are available, including the time histories of the GPU baseplate, bus, and radiator temperatures. - The thermal pad and heat pipe parameters of the whole-satellite model are converted into R_{JC} and R_{CR} according to T5, and the conversion method and sources are recorded.		
Input Data	- Two examples: 12 V100 GPUs and 12 A100 GPUs. - The same orbit, attitude, and load power histories as the finite-element model.		
Procedure	- Assemble the three components J, C, and R from the whole-satellite model parameters. - Apply the same power history and run five orbits. - Take the fourth and fifth orbits and compare the mean GPU baseplate temperature and the mean radiator temperature.		
Expected Results	The chain from the computing node through the cold plate to the radiator reproduces the temperature difference between the computing equipment and the radiator. In the same comparison, the single-node model is 12 to 15 °C lower than the GPU baseplate.		
Acceptance Criteria	In both the fourth and fifth orbits, the mean temperature differences of the GPU baseplate and of the radiator shall not exceed 5 K.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	Two whole-satellite cases with 12 V100 and 12 A100 devices were run, comparing fourth- and fifth-orbit means. Computing-node temperatures exceed FE GPU baseplate temperatures by 7.9 to 12.5 K; radiator temperatures exceed FE values by 5.9 to 9.2 K. Several values exceed 5 K and the case fails. The single-node controls are 12.46 K and 15.06 K below the FE baseplate; the latter exceeds the expected 12 to 15 K interval. The FE energy-budget remainder beyond four main radiator faces includes surfaces belonging to several nodes, which available exports cannot separate. It cannot be used in full to correct node temperatures or validate chain resistance. Assigning the bus to node D does not remove the discrepancy. Test program tests/test_fe_004_whole_satellite_chain.py recorded 46 checks, 36 passed, 10 failed. Evidence file: tests/results/FE-004.json.		
Recorded By		Test Conclusion	[] Pass [X] Fail [] Not Executed
Anomalies	The case does not meet all recorded requirements. The measured exceedances and their evidence limits are stated above and in the comparison tables. Failed check count: 10. See tests/results/FE-004.json for each expected value, actual value and verdict.		

Table 22 and Figure 3 compare the mean temperatures of orbits 4 and 5; Table 23 gives the FE energy-budget remainder.

Table 22. FE-004 mean temperatures of orbits 4 and 5

Case	Orb.	GPU baseplate FE °C	Node J module °C	Diff. K	Radiator FE °C	Radiator module °C	Diff. K
12 × V100	4	58.77	66.69	7.92	39.81	45.67	5.86
12 × V100	5	59.61	68.13	8.52	40.60	46.86	6.26
12 × A100	4	78.33	90.08	11.75	53.91	62.52	8.61
12 × A100	5	78.15	90.67	12.52	53.94	63.09	9.15

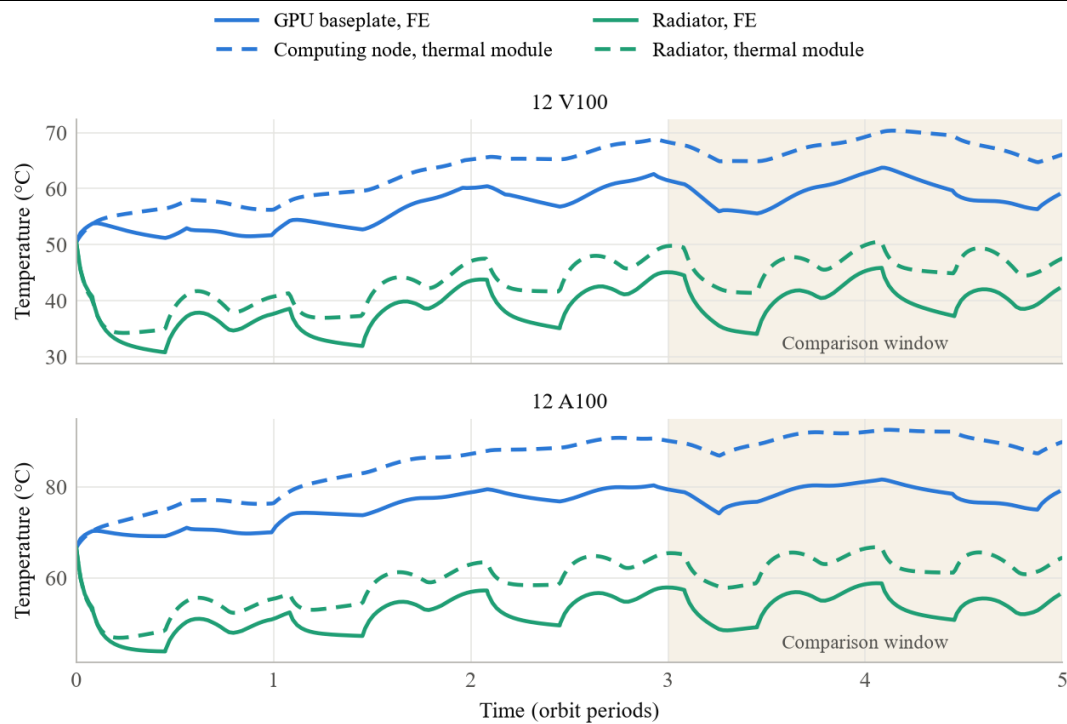


Figure 3. FE-004 GPU baseplate, computing node and radiator temperature histories

Table 23. FE-004 FE energy-budget remainder

Case	Orb.	Sources W	Four main faces and storage W	Remainder W	Share %
12 × V100	4	2352	2120	207	8.8
12 × V100	5	2208	2126	228	10.3
12 × A100	4	3108	2683	375	12.1
12 × A100	5	2940	2662	386	13.1

The remainder includes several surface contributions outside the four main radiator faces, including radiator edges and other components. Available exports cannot separate these

contributions. The entire remainder cannot be assigned to the computing node, used to correct temperatures, or used to infer and validate the chain resistance.

3.6 Numerical Calculation and End-to-End Operation

3.6.1 Test Project 14: Numerical Calculation and Eclipse Boundaries

This project verifies the numerical settings of the SDTwin thermal module specified in Chapter 8 of the design report.

Table 24. NI-001 - Numerical Calculation and Eclipse Boundaries Test Case

Case ID	NI-001	Case Name	Numerical Calculation and Eclipse Boundaries
Test Objective	Verify the adaptive ODE method, tolerances, maximum step, handling of eclipse and load-switching boundaries, and output sampling.		
Design Basis	Chapter 8, Table 8; Chapter 7		
Preconditions	The mean environment case of FE-001 can be run.		
Input Data	- Integration methods RK45 and Radau. - Relative tolerances 1×10^{-6} and 1×10^{-8}; temperature absolute tolerances 1×10^{-6} K and 1×10^{-8} K; maximum steps 60 s and 10 s. - Output intervals 10 s and 60 s; one load-switching instant; one integration step that must be rejected.		
Procedure	- Run each combination once and compare the last-orbit temperatures. - Check that at eclipse entry and exit and at load switching the integration is split at the boundary and is not averaged across it. - Check that the output samples the accepted solution at the required times, independently of the internal integration step. - Check that a rejected integration step writes no component temperatures.		
Expected Results	- Results converge as the tolerances are tightened, and RK45 and Radau agree. - Boundary handling and output sampling conform to Chapter 8.		
Acceptance Criteria	After the tolerances are tightened by a factor of 100, the change in the last-orbit temperatures shall not exceed 0.01 K. The difference between RK45 and Radau shall not exceed 0.01 K. The error in eclipse entry and exit times shall not exceed 1 s. Output sampling shall not change with the internal step.		
Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	RK45 and Radau, two relative and absolute tolerances, two maximum steps and two output intervals form a 32-setting matrix, supplemented by direct-environment and rejected-step runs. Tightening tolerances 100-fold changes temperature by at most 5.53×10^{-4} K; the largest method difference is 5.81×10^{-4} K, compared with 0.01 K. The largest eclipse-location error is 3.64×10^{-12} s. Fixed output grids, segmentation and independent reference solutions are recorded for the 32 settings. Solver statistics and trial logs check exclusion of rejected states from output. Test program tests/test_ni_001_numerics_eclipse.py recorded 33 checks, 33 passed. Evidence file: tests/results/NI-001.json.		

Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.6.2 Test Project 15: End-to-End Operation and Result Archive

This project verifies the joint operation of the SDTwin thermal module with Orbit and Power, following the workflow in Chapter 7 of the design report.

Table 25. NI-002 - End-to-End Operation and Result Archive Test Case

Case ID	NI-002	Case Name	End-to-End Operation and Result Archive
Test Objective	Verify the synchronized inputs, constraint-event handling, temperature ports, single ownership of component temperatures, and result archiving.		
Design Basis	Section 3.3; Section 6.3; Chapter 7; Section 9.4		
Preconditions	The Orbit and Power modules are available, and the asset revisions, initial values, and environmental data are frozen.		
Input Data	- A 24 h scenario containing eclipses, load switching, and battery constraint events. - One trial in which Power returns valid false and one trial in which Power returns event_required true.		
Procedure	- Run the scenario in the calling order of Section 6.3 and save all outputs. - Check that an invalid Power result is reported as an error and stops that evaluation, and that when event_required is true the calculation first returns to the constraint boundary, updates the discrete load state according to the declared rule, and then recalculates the powers and temperature derivatives. - Check that temperatures are saved only after an integration step is accepted, that the computing domain reads the temperature of <i>J</i>, and that Power reads the temperature of <i>B</i> for its next trial. - Check that, with the thermal module enabled, power_considering_thermal of Orbit no longer updates the same solar array temperature, and check that the thermal module adds no power-off or throttling rules. - Repeat the run with identical inputs, compare the two results, check the archive fields, and record the number of Power root evaluations and rejected trials.		
Expected Results	- Events and invalid results are handled according to Chapter 7. - Each component temperature has only one update source. - The two runs give identical results, and the archive is complete.		
Acceptance Criteria	The two runs shall be bitwise identical, or their difference shall not exceed 1×10^{-12} . The archive shall contain run_id, the UTC epoch, time_s, component identifiers and state order, temperatures, the four Power ports, heat flows, solver settings, and parameter revisions. All checks of steps 2 to 4 shall pass.		
Case Designer		Executor / Date	Automated test, 2026-10-06

Actual Results	Orbit, earth_flux, the Power stand-in and the six-node thermal model were coupled for 24 h with a 55° solar-array incidence angle, a 60 W base request and a 900 s request ramp. The archive contains 1441 output samples. Two identical-input runs compare 106826 archive leaves, including 89161 numeric leaves, with 0 differences. A separate fault-injection run checks invalid-result stopping. Injected supply loss and a physical battery constraint separately exercise location, disconnection and restart. The test driver writes the pre-run parameter record. Power validation temperature agrees with the thermal initial value; runtime battery temperature comes from the current thermal state. Test program tests/test_ni_002_end_to_end.py recorded 36 checks, 36 passed. Evidence file: tests/results/NI-002.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

3.6.3 Test Project 16: Asset and Scene Assembly

This project verifies the function of the SDTwin thermal module that assembles the thermal model from SimReady assets and scene configuration.

Table 26. NI-003 - Asset and Scene Assembly Test Case

Case ID	NI-003	Case Name	Asset and Scene Assembly
Test Objective	Verify the Table 9 records, instance assignments, geometry units, overrides, the shared temperature node, independent instances, and single counting of mass.		
Design Basis	Section 9.3, Table 9; Section 9.4		
Preconditions	The assets of the seven instances under Sat01 have asset_id, asset_version, geometry_uri, domain, model_id, parameter_set, ports, and provenance registered.		
Input Data	- SolarArray01, Battery01, Controller01, PDU01, Compute01, ColdPlate01, and Radiator01, with geometry paths under /World/Sat01 named by instance identifier. - One display-scaled asset; one set of parameter_overrides; two instances created from the same asset; one asset with an unregistered model_id.		
Procedure	- Assemble the scene, read the areas, masses, and thermal capacitances, and compare them with hand calculations from the physical dimensions of the assets. - Check that the display-scaled bounding box is not taken as the physical dimensions. - Check that parameter_overrides take precedence over the asset parameter set, and that the affected thermal capacitances, areas, and thermal resistances are recalculated after a change of material, geometry, or installation. - Check that Controller01 and PDU01 share <i>D</i>, that the battery thermal connection BR leads to the radiator, that Power and Thermal use the same component instance identifiers, and that material mass is counted once. - Check that the two instances have independent temperatures and battery states, that the unregistered model_id is rejected and no arbitrary code is executed from USD, and that the photovoltaic efficiency does not exceed the absorptivity of the same surface.		
Expected Results	- Physical parameters are determined only by physical dimensions, materials, and installation. - Instance assignments and the shared temperature node are correct.		
Acceptance Criteria	The relative errors of areas and masses shall not exceed 0.1%. All checks of steps 2 to 5 shall pass.		

Case Designer		Executor / Date	Automated test, 2026-10-06
Actual Results	SimReady assets, USD physical dimensions, display scale, material masses, scene overrides and model registration were checked. The largest relative errors of area, mass, capacitance and resistance are 0, 1.43×10^{-16}, 1.63×10^{-16} and 0. Four material portions use size, density and solid fraction; the others use declared masses. Two satellite scenes run for 5600 s each, with identical repeat and separate-process archives. Example lithium fractions were explicitly corrected before execution to conserve the test cell inventory; runtime does not replace scene values. Parameter provenance retains effective density and mass sources after overrides, and text embedded in USD is not executed. Test program tests/test_ni_003_scene_assembly.py recorded 63 checks, 63 passed. Evidence file: tests/results/NI-003.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	None		

4 Test Content Sufficiency Analysis

The 16 cases map to the design report chapter by chapter, as listed in Table 27: Equations T1 to T5, the four data objects and five functions of Chapter 5, the interface conventions of Chapter 6, the run procedure of Chapter 7, the numerical settings of Chapter 8 and the asset assembly and result archiving of Chapter 9 all have cases, and so do the four requirements TH-01 to TH-04, traced in Table 1.

Table 27. Design Report Sections and Test Cases

Design report section	Verified content	Test cases
2 Requirements and Design Objectives	Scene model assembly and component simulation requirements	PA-001, NI-003, HT-002, FE-001 to FE-004
3.1 System architecture	Components and heat-transfer connections in Figure 1	HT-002, FE-004
3.2 Package decomposition	Files and exports of the thermal package	DM-001
3.3 Simulation workflow	Calculation sequence and temperature ports in Figure 2	NI-002
4.1 Overall Heat Balance	Equation T1	HT-002
4.2 Heat Transfer Between Components	Equation T2	HT-001
4.3 Component Temperatures	Equation T3 and the battery example	HT-002, HT-003, EC-001, FE-001 to FE-004
4.4 Environmental Absorption and Surface Radiation	Equation T4 and Table 5	EN-002, EN-003, FE-001, FE-002
4.5 Thermal Capacitance and Resistance	Equation T5 and the example	PA-001, FE-002, FE-003, FE-004

Design report section	Verified content	Test cases
5.1 Common data models	Four data objects in Table 6	DM-001
5.2 Thermal parameter assembly	assemble_thermal_parameters	PA-001
5.3 Surface environment preparation	prepare_surface_environment and earth_flux	EN-001, EN-003
5.4 Surface absorption and emission	calculate_surface_heat	EN-002
5.5 Component heat-transfer calculation	calculate_heat_flows	HT-001
5.6 Temperature derivative calculation	thermal_derivative and the Power coupling	HT-002, EC-001
6.1 General contracts	Time, array order, power and temperature fields, Orbit outputs	DM-001, EN-001, EC-001
6.2 Primary calling interfaces	Five calls in Table 7	PA-001, EN-001, EN-002, HT-001, HT-002
6.3 Minimal end-to-end call	Calling order and integration-step acceptance	NI-002
7 End-to-End Operational Workflow	Joint trials, constraint events, and temperature ports	NI-002
8 Numerical Design and Performance	Numerical methods and controls in Table 8	HT-003, NI-001
9.1 Runtime; 9.2 Installation and interpreter configuration	Interpreter, dependency, and source revisions	Test conditions in Chapter 5
9.3 SimReady assets and scene assembly	Records and instances in Table 9	NI-003
9.4 Parameter configuration and result archive	Provenance, initial values, simplifications, and archive	PA-001, EN-003, FE-001, NI-002
10 Limitations and Planned Evolution	Lumped temperatures, radiation between components, equivalent resistances	FE-002 to FE-004 record the magnitude of the limitations
11 Requirements Traceability Matrix	TH-01 to TH-04	All test cases in Table 1
Appendix A Public API Inventory	Proposed public names	DM-001
Appendix B Model-Selection Guidance	Finite-element identification of equivalent parameters, and local hot spots	FE-003, FE-004

Table 28 lists the reference and execution status of each comparison. References are design report examples, hand calculations, analytic solutions, independent numerical calculations and finite element results; no reference calculation uses thermal module code.

Table 28. Comparisons, References and Execution Status

No.	Comparison	Reference	Available data	Execution status
1	Fields, dimensions, and order of the data objects	Table 6 of the design report	None	Pass, DM-001
2	Thermal capacitances and resistances	Hand calculation	Example in Section 4.5 of the design report; ISS model parameter table	Pass, PA-001
3	Incidence cosine and eclipse	Independent vector calculation; Orbit output	None	Pass, EN-001
4	Absorbed and emitted powers	Hand calculation	Example in Section 4.4 of the design report	Pass, EN-002
5	Earth albedo and infrared irradiance	Analytical view factor; numerical integration; finite-element orbital heat loads	Solved ISS finite-element model	Fail, EN-003
6	Connection heat flows and temperature derivatives	Hand calculation; overall heat balance T1	Battery example in Section 4.3 of the design report	Pass, HT-001, HT-002
7	Temperature integration	Analytical solutions	None	Pass, HT-003
8	Electrical/thermal coupling	Constructed examples; test power program	None	Pass, EC-001
9	Solar array temperature	ISS finite-element results for the US solar array wings	Third-orbit results of three cases	Fail, FE-001
10	Radiator temperature	ISS finite-element results for the EATCS radiators	Third-orbit results of two loops in three cases	Pass, FE-002
11	Computing node temperature	ISS finite-element results for MBSU, DDCU, and IEA	Third-orbit results of the mean environment case	Pass, FE-003
12	Chain from the computing node to the radiator	Whole-satellite finite-element results	Five-orbit results of two examples	Fail, FE-004
13	Numerical settings	Own results with tightened tolerances; analytical solutions	None	Pass, NI-001
14	End-to-end workflow and archive	Workflow in Chapter 7 of the design report; repeated runs	None	Pass, NI-002

No.	Comparison	Reference	Available data	Execution status
15	Asset and scene assembly	Physical dimensions of the assets; Table 9 of the design report	None	Pass, NI-003

The finite element comparisons cover the solar array, the common radiator, the computing node and the cold plate, and the JC and CR connections. The battery and power equipment and the SR, BR and DR connections have no finite element reference and are verified with design report examples, hand calculations, analytic solutions and the overall heat balance. The end-to-end cases use the Power stand-in; the battery electrochemical model and device parameters still require separate Power-module validation.

The results also quantify the limitations listed in Chapter 10 of the design report: at the same instant the coldest radiator panel is 18.8 to 20.6 K below the radiator mean; FE loop B is about 4.7 K warmer than loop A in the design hot case, against about 1.6 K in the module. The contributions of wing obstruction and along-flow gradients are not isolated. The whole-satellite energy budget has a roughly 9% to 13% remainder beyond four main radiator faces and storage. Available exports cannot allocate it to individual nodes or establish the complete cause of FE-004 exceedances.

Every case has an execution record; 3 cases do not meet acceptance. Their discrepancies require correction and repeat testing before reassessment. This report retains the measured results.

5 Test Conditions and Requirements

- Record the thermal module source revision, the Python and dependency revisions, the operating system, and the processor, consistent with the runtime environment in Section 9.1 of the design report.
- Record the UTC epoch, frames, and component state order; inside the module, temperatures are in K, time in s, and powers and heat transfer in W.
- According to Section 9.4 of the design report, save the resolved parameters, provenance, initial values, and environmental configuration before startup; for cases with environmental terms disabled, record the simplification.
- Freeze the asset revisions, parameter sets, environmental cases, and finite-element reference files, and record their hashes.
- Run reference calculations independently; the design report examples, hand-calculation tables, analytical solutions, and finite-element results do not use thermal module code.
- For ISS temperature acceptance, take the third orbit, compute time-weighted statistics over the exact orbital period, and compare the mean temperatures of the same part. The planet-refinement experiment separately compares thermal loads at common sampling instants.

The execution environment was Windows 11 Home, Python 3.11.9, NumPy 2.4.6, SciPy 1.17.1, Astropy 8.0.1, pytest 9.1.1, the Orbit package `ntu_space_dynamics` 0.1.0 and OpenUSD 0.26.8; finite element thermal loads were exported with COMSOL 6.3 and MPh. All cases are run from the Thermal directory with `.venv/Scripts/python.exe -m pytest tests -q`. The final suite log is `tests/results/pytest_verified_run.txt`; the EN-003 rerun after COMSOL refinement is `tests/results/pytest_en003_final.txt`. The final `verification_manifest.json` records case execution times, completeness and file hashes. The checks, key values, result text and anomalies of each case are retained in its named result file under `tests/results`.

6 Test Data

Test inputs and results are retained in the project directory. The evidence manifest records file hashes; data records state origin, validity range, time and frame conventions.

Table 29. Test Data Inventory

No.	Test Data	Data Type	Form	Source
1	SDTwin Thermal Software Module Design Report	Basis	DOCX	Thermal/SDTwin_Thermal_Design_Report_EN.docx
2	SDTwin Power Software Module Design Report	Basis	DOCX	Power/SDTwin_Power_Design_Report_EN.docx
3	Thermal module and test support programs	Item under test	Python	Thermal/thermal and Thermal/sdtwin_sim; interface conventions in Thermal/IMPLEMENTATION.md
4	Test programs, hand-calculation tables, and test case data	Program and reference	Python, JSON, USD	Test programs and the data directory under Thermal/tests
5	ISS model parameters	Input	Python	Thermal/iss_fem/model/iss_spec.py
6	Finite-element time histories of the three cases	Reference	CSV	series.csv in each case directory under Thermal/iss_fem/out
7	Finite-element part statistics and loop heat breakdown	Reference	CSV	items.csv and loop_breakdown.csv in the same directories
8	Solved finite-element model and exported heat loads	Reference	MPH, JSON	Thermal/iss_fem/out/comsol; exported results in Thermal/tests/data/en_003
9	Whole-satellite finite-element results	Reference	CSV, JSON	space-compute-demo/tools/comsol_benchmark/full_twin/out
10	Execution evidence	Evidence	JSON, TXT	Case results, time histories, pytest_verified_run.txt, pytest_en003_final.txt and verification_manifest.json under Thermal/tests/results

7 Test Summary

All 16 cases were executed: 13 passed and 3 failed. Each status was evaluated against the acceptance criteria with retained execution evidence.

Table 30. Thermal Module Test Summary

No.	Case ID	Function	Verification Focus	Status	Result / Evidence
1	DM-001	ThermalParameters, ThermalState, ThermalInputs, ThermalEvaluation	Fields, dimensions, order, read-only parameters, and state separation	Pass	All 217 checks passed; the four data objects match Table 6 and all 63 invalid constructions are rejected. Evidence: tests/results/DM-001.json
2	PA-001	assemble_thermal_parameters	T5 capacitances and resistances, overrides, duplicate assignment, provenance	Pass	All 114 checks passed; largest relative error of capacitances and resistances 1.0×10^{-16} ; the examples give 1000 J/K, 108.7 kJ/K and $7.2 \text{ kJ} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ as required. Evidence: tests/results/PA-001.json
3	EN-001	prepare_surface_environment	Sun direction, quaternion rotation, incidence cosine, eclipse counted once	Pass	All 88 checks passed; largest $\cos_{\text{incidence}}$ error 7.8×10^{-16} ; G identical to the Orbit output. Evidence: tests/results/EN-001.json
4	EN-002	calculate_surface_heat	T4 direct, albedo, and infrared absorption and surface emission	Pass	All 62 checks passed; largest relative error 1.9×10^{-16} ; the Section 4.4 example gives 1600 W. Evidence: tests/results/EN-002.json
5	EN-003	earth_flux environment input	Earth-facing view factor, albedo and infrared irradiance, missing fields	Fail	17 of 89 checks failed; weak-albedo relative errors exceed 1%; the original FE comparison does not fully meet 3%; three-case refinement results are tabulated separately. Evidence: tests/results/EN-003.json
6	HT-001	calculate_heat_flows	T2 signed heat flows, path order	Pass	All 91 checks passed; largest relative error of heat flows 9.7×10^{-17} ; the battery example gives 6 W and -6 W . Evidence: tests/results/HT-001.json
7	HT-002	thermal_derivative	T3 temperature derivatives, T1 overall heat balance	Pass	All 21 checks passed; battery example 0.004 K/s; largest T1 residual ratio 8.7×10^{-16} . Evidence: tests/results/HT-002.json
8	HT-003	thermal_derivative and the numerical solver	Exponential response, series-resistance steady state, radiative equilibrium	Pass	All 27 checks passed; largest step-response error 0.0000018 K; both steady cases within 0.001 K. Evidence: tests/results/HT-003.json
9	EC-001	ThermalInputs and thermal_derivative	Four power ports, signs and upper limit, single ownership of battery temperature	Pass	All 83 checks passed; each power enters only its component; largest relative error of derivative changes 6.0×10^{-16} ; battery temperature updated only by the thermal module. Evidence: tests/results/EC-001.json

No.	Case ID	Function	Verification Focus	Status	Result / Evidence
10	FE-001	Solar array <i>S</i>	Environmental absorption, subtraction of actual electrical output, two-sided surface emission	Fail	3 of 71 checks failed; temperature statistics within 1.69 K; cold and mean pointwise maxima 9.18 K and 12.09 K exceed 3 K. Evidence: tests/results/FE-001.json
11	FE-002	Common radiator <i>R</i>	Branch heat inputs, environmental absorption, two-sided surface emission, thermal capacitance	Pass	All 54 checks passed; largest absolute third-orbit mean difference 3.39 K against 5 K. Evidence: tests/results/FE-002.json
12	FE-003	Computing node <i>J</i> and cold plate <i>C</i>	JC resistance composition, mean temperature of uniformly heated equipment	Pass	All 63 checks passed; largest mean temperature difference of 14 units 0.25 K against 1 K. Evidence: tests/results/FE-003.json
13	FE-004	Computing node, cold plate, and radiator chain	Complete temperature drop from the computing equipment to the radiator	Fail	10 of 46 checks failed; computing node 7.9 to 12.5 K above the GPU baseplate and radiator 5.9 to 9.2 K above the FE against 5 K; effects of different surface coverage have not been isolated. Evidence: tests/results/FE-004.json
14	NI-001	Numerical solver settings	Tolerance convergence, method agreement, eclipse and load-switching boundaries, output sampling	Pass	All 33 checks passed; change 5.5×10^{-4} K for 100 times tighter tolerances, RK45 and Radau within 5.8×10^{-4} K, eclipse times within 3.6×10^{-12} s. Evidence: tests/results/NI-001.json
15	NI-002	Joint operation and result archive	Event handling, temperature ports, single temperature source, archive and reproducibility	Pass	All 36 checks passed; 24 h coupled run completed, two runs identical bit for bit, events and invalid results handled as in Chapter 7. Evidence: tests/results/NI-002.json
16	NI-003	SimReady assets and scene configuration	Table 9 records, geometry units, overrides, shared node, independent instances	Pass	All 63 checks passed; largest relative error of areas and masses 1.4×10^{-16} ; all checks of steps 2 to 5 passed. Evidence: tests/results/NI-003.json

The failed cases are EN-003, FE-001, FE-004. EN-003 retains original FE exceedances and separately records weak-flux integration accuracy and planet-refinement results; diagnosis does not replace acceptance. FE-001 meets the temperature-statistic criterion but fails simultaneous curve comparison; eclipse treatment and time resolution require further matched checks. FE-004 uses different radiating-surface coverage in the two models. The available energy budget cannot isolate each node contribution and does not establish validation of resistance or temperature. Passing formula, interface or integration checks does not replace FE temperature and environmental-load acceptance. Supported claims are limited to each case record.

Overall Conclusion: [] Pass [X] Fail [] Not Executed

On-site Test Expert Signature: _____ Date: _____